



Building_Energy_Modeling_Energy_Efficiency_Measures_Framework

EEM_Journal — One-Page Framework Explainer

What is EEM_Journal?

EEM_Journal is a retrofit bundle designed to quantify the marginal contribution of envelope, ASHP, and HPWH upgrades for a journal paper chapter. It applies three cumulative EEM tiers (EEM1 → EEM2 → EEM3) to a 64-building Canadian cohort (32 per climate) so each heat-pump measure layer can be attributed separately in an EUI ladder.

EEM_Journal is distinct from the dormant 33-measure EEM_MVP_ADV production stack. It focuses exclusively on heat-pump upgrades; non-HP measures active in earlier versions (HRV/ERV, supply-fan SFP tune, DHW demand-side and distribution) were intentionally cut in v6 to keep HVAC-tier = ASHP and DHW-tier = HPWH attribution clean (see §v6 X-cuts and deferrals).

Scenarios

Scenario tag	Short name	Measures applied
EEM_J_DEFAULT	Default	Unmodified baseline IDF; native PV only
EEM_J_ENVELOPE	EEM1	Envelope package (see below)
EEM_J_ENV_HVAC	EEM2	EEM1 + ASHP nameplate uplift + CCHP curves applied + inverter cooling curves + per-zone PTHP + defrost OnDemand + supplemental heater lockout + desuperheater
EEM_J_ENV_HVAC_DHW	EEM3	EEM2 + HPWH CO ₂ retrofit (Stratified, COP 4.0, −29 °C lockout, outdoor-air) + MURB swing-tank + SF preheat-tank
EEM_J_ENV_HVAC_DHW_EEM4	EEM4	EEM3 + automated blinds + daylight dimming + LED LPD ×0.55 + occupancy-sensor schedule trim + ENERGY STAR plug loads + smart power management + APS phantom cut + HEMS peak trim + gas-appliance retrofit (BTAP residential only)

Each scenario is cumulative: EEM4 ⊂ EEM3 ⊂ EEM2 ⊂ EEM1 as strict supersets.

What each scenario contains

EEM1 — Envelope

Measure	Detail
Envelope constructions	HPENV high-performance opaque assemblies (wall, roof, floor) from <code>constructionSet_baseline.idf</code>
Triple-glazed windows	U=0.85 W/m²K, SHGC=0.40 — replaces all glazing in baseline
Foundation / slab insulation	Climate-zone-gated: applied in CZ6 (Montréal) and CZ7A (Calgary); skipped for warmer zones
Infiltration ×0.25	Design-flow air leakage reduced to Passive House target (×0.25); was ×0.5 in v2

References:

- **Envelope constructions:** NRCan/CanmetENERGY (2023) — *Deep Energy Retrofits for Canadian Buildings* (71% natural-gas heating reduction; up to 80% total heating demand reduction); Lavigne et al., NRCan BTAP (2018–2022) — 15–30% whole-building EUI reduction on NECB-compliant CZ6/CZ7A baselines.
- **Triple-glazed windows:** PNNL/LBNL (2022) — *Thermal Performance of Triple-Pane Windows in Cold Climates* (16% heating reduction; 3–7% total site EUI reduction); NRC/BTAP (2021) — *NECB 2017/2020 Tiered Code Analysis* (50–60% savings over NECB 2017 Tier 3/4).
- **Foundation / slab insulation:** NRCan/CanmetENERGY (2023) — exterior insulation + air sealing; corroborated by NRC/BTAP (2021).
- **Infiltration ×0.25:** Hosseini M. et al. (2017) — *Comparison of building energy codes* (validates thermal bridging + infiltration savings in cold climates).

EEM1 is frozen at v4 spec for v6. No EEM1 changes were made in v6.

EEM2 — Envelope + ASHP

All EEM1 measures plus:

Measure	Function	Detail
ASHP nameplate uplift (E2-C)	<code>_convert_heating_to_ashp</code>	SF DX coils → COP 4.5 heating / EER 17 cooling [v6 X1: renamed from "GSHP option"]
CCHP curves applied (E2-D)	<code>_build_cchp_curves</code>	Biquadratic Cap-FT / EIR-FT curves wired to all DX heating coils; COP=2.5 floor; −25 °C compressor lockout (C1). Note: in v2/v3 these curves were <i>built but not applied</i> ; v6 fully activates them.
Inverter cooling curves (E2-I)	<code>_apply_inverter_cooling_curves</code>	SEER 22 / EER 14 Cap-FT and EIR-FT curves on SF cooling DX; PLF = 1.0 flat [C4 — eliminates 7–8% cycling tax]
Per-zone PTHP (E2-J Branch A)	<code>_convert_sf_central_to_ductless_pthp</code>	Central DX deleted; per-zone PTHP units on Detached/Attached SF archetypes; AFN → <code>MultizoneWithoutDistribution</code>
Defrost OnDemand (C3)	<code>_apply_defrost</code>	<code>Defrost_Control = 'OnDemand'</code> ; flat <code>EEM2_ASHP_DefrostEIR_Flat</code> curve [v6 correctness fix — silently skipped in v4/v5]
Supplemental heater lockout (C2/G4)	<code>_apply_lockout_temps</code>	CZ6 (Montréal): −22 °C lockout; CZ7A (Calgary): −27 °C lockout [v6 fix — was +4/+5 °C in v4/v5, firing backup resistance during shoulder season]
Desuperheater HP→DHW (E2-N)	<code>apply_desuperheater_dhw</code>	<code>Coil:WaterHeating:Desuperheater</code> on SF cooling coil; $\eta_{\text{recovery}} = 0.30$; routes to A5 preheat tank

References:

- **ASHP nameplate uplift (E2-C):** IEA HPT Annex 52 (2018–2021) — average HP COPs 4.0; optimized systems up to 7.2; IGSHPA (2024) — *Cold Week Case Study*: COP 4.5 during peak January cold snap.
- **CCHP curves (E2-D):** Mendon et al., PNNL (2025) — *DOE Cold Climate Heat Pump Challenge*: median COPs 1.9 at −15 °C and 1.8 at −25 °C; CCHT/NRCan (2018) — 20–30% COP drop during extreme cold snaps, validates defrost-EIR curve.
- **Inverter cooling curves (E2-I):** BA-PIRC/FSEC (2015/2018) — 22–33% cooling savings for SEER 22 vs SEER 13/14; NREL (2022–2024) — *Basalt Vista Study*: validated SEER 22/HSPF 10.2 in cold climates.

- **Per-zone PTHP (E2-J Branch A):** Purdue University/DOE — *Thermal Distribution Efficiency*: 20–30% duct tax eliminated by per-zone topology in tightened envelopes.
- **Defrost OnDemand (C3):** CCHT/NRCAN (2018) — defrost cycles cause 20–30% COP drop in humid cold climates; OnDemand adds 5–10% HSPF at no risk.
- **Supplemental heater lockout (C2/G4):** Mollier et al., NRCAN (2023) — ccASHPs maintain COP > 1.5 below –20 °C, eliminating need for backup above design OAT.
- **Desuperheater HP→DHW (E2-N):** ASHRAE 90.1-2022 §6.5.6.2 — refrigeration heat reclaim mandate; NREL TP-7A40-67762 — $\eta_{\text{recovery}} = 0.30$ measured on split-system DX desuperheaters.

EEM3 — Envelope + ASHP + HPWH

All EEM2 measures plus:

Measure	Function	Gate	Detail
HPWH CO ₂ retrofit (E3-F)	<code>apply_hpwh_retrofit_journal</code>	Broad	<code>WaterHeater:Mixed</code> → <code>HeatPump:PumpedCondenser</code> transcritical CO ₂ ; OutdoorAirOnly; rated COP 4.0 (Phase L — Sanden GAU / Mitsubishi Q-Ton class); –29 °C lockout
HPWH tank → Stratified (C7)	<code>_convert_mixed_tank_to_stratified</code>	Broad	All HPWH tanks: Mixed → Stratified (6 nodes); gas-cooler inlet at bottom node [v6 correctness fix; Phase P labeling fix pending — <code>EndUse_Subcategory</code> correction; see Phase P note below]
HPWH outdoor-air intake (G2)	—	CZ≥4	<code>AirInlet_Configuration</code> = <code>OutdoorAirOnly</code> for CZ6/CZ7A; warm-CZ <code>ZoneAirOnly</code> branch coded but untested in v6
HPWH compressor cutoff (C6/G3)	—	Broad	–29 °C lockout on all HPWH wrappers including alt-path [v6 fix — alt-path was +5 °C in v4/v5, locking out compressor all winter in CZ6/CZ7A]
HPWH off-peak boost (E3-G)	<code>apply_hpwh_setpoint_reset</code>	Broad	02:00–06:00 setpoint 65 °C + tempering valve 50 °C
HPWH tank jacket R-25 (E3-J)	<code>apply_hpwh_tank_jacket</code>	Broad	Off/On cycle-loss coefficients ×0.64 (R-16 → R-25); UA floor assert ≥ 0.3 W/K (C11)
MURB swing-tank topology (A1+A2)	<code>apply_swing_tank</code>	MURB	50-gal Mixed swing tank + 75 W recirc parasitic inserted between recirc return and CO ₂ HPWH supply
SCWH integrated HP (A3) [†]	—	Hotel/Hospital/SuperTall predicate	Code-only in v6: predicate evaluates False on all 4 residential archetypes; deferred to v7 commercial round
Preheat-tank topology (A5)	<code>apply_preheat_tank</code>	SF	30-gal Mixed preheat tank upstream of SF HPWH; enables E2-N desuperheater coexistence (avoids v22.1 Fatal)
Hydronic combi HP (A6/I4a) [†]	—	Radiant-loop commercial predicate	Code-only in v6.y: predicate gates to <code>ZoneHVAC:LowTemperatureRadiant:*</code> + non-DHW hot-water PlantLoop; no v6.y residential sim affected; deferred to commercial dataset round

References:

- **HPWH CO₂ retrofit (E3-F):** Energy350 BC Study (2018) — CO₂ split systems reliable to –29 °C; COP > 2.50 at –10 °C; NEEA/WSU (2015/2018) — seasonal COP 2.7–3.4, operation to –25 °C; Goudarzi et al. (2021) — 30–50% DHW reduction vs R-134a in cold-climate residential; NREL/PNNL Split-System HPWH (2022) — outdoor-air intake validates –29 °C lockout.
- **HPWH tank → Stratified (C7):** NREL (2022) — Stratified CO₂ HPWH achieves +5–15% COP vs Mixed tank due to correct cold-inlet temperature at bottom node.

- **HPWH compressor cutoff (C6/G3):** Energy350 BC Study (2018) — CO₂ class operates to −29 °C; NEEA/WSU (2018) — validates operation to −25 °C.
- **HPWH outdoor-air intake (G2):** NREL/PNNL Split-System HPWH (2022) — outdoor-air split eliminates zone-air theft and false COP credit.
- **HPWH off-peak boost (E3-G):** PG&E WatterSaver Program (2024) — Advanced Load Up to 65 °C, ~500 Wh shifted per unit; PNNL/FSEC (2021–2023) — ~0.5 kW peak shaving per unit.
- **HPWH tank jacket R-25 (E3-J):** ASHRAE Handbook HVAC Systems (2020) Ch. 50 — tank standby loss scaling by 1/R.
- **MURB swing-tank (A1+A2):** NEEA/WSU CO₂ HPWH Field Study — Anderson et al. (2017/2018): swing-tank topology required for MURB centralized CO₂ HPWH; isolates warm recirc return from gas cooler; field-validated ~12 kWh/m² DHW; recirc pumps measured at 40–75 W.
- **SCWH integrated HP (A3):** DOE (2021) — >70% DHW savings during cooling season for Hotel/Hospital with SCWH mode.
- **Preheat-tank topology (A5):** NREL TP-7A40-67762 — preheat-tank topology enables split-system desuperheater without shared condenser-sink conflict.
- **Hydronic combi HP (A6/I4a):** IEA HPT Annex 56 (2023) — *Combi heat pumps for low-energy buildings*: predicate-gated to commercial radiant-loop archetypes.

[†] Not exercised on the v6 residential fleet (predicate evaluates False on all four residential archetypes); kept as scaffolding for the v7 commercial round.

EEM4 — Envelope + ASHP + HPWH + Cooling/Lighting/Equipment

All EEM3 measures plus:

Cooling-side (shading + daylighting)

Code	Measure	EnergyPlus objects	Detail
C4-1	Automated interior venetian blinds	WindowMaterial:Blind + WindowShadingControl (OnIfHighSolarOnWindow , 200 W/m ² + OAT ≥ 15 °C; BlockBeamSolar slat angle)	Applied on all exterior windows with WWR ≥ 0.10; skipped on skylights, clerestories Z > 2.5 m, interior windows, north-facing in CZ6/CZ7A; skipped entirely on Laboratory, TallBuilding, SuperTallBuilding (blank-name fenestration E+ v22.1 fatal)
C4-3	Continuous daylighting dimming — perimeter zones	Daylighting:Controls ContinuousOff ; setpoint 300 lux residential / 500 lux commercial (IES RP-1 / NECB); reference point 1.5 m inward (residential) or 3.0 m inward (commercial); fraction_controlled ≥ 0.10 gate	Skipped on zones with continuous-operation schedules (Min_Fraction > 0.40 → life-safety / process); skipped on surgical/exam/lab/kitchen/fume zones

Lighting-side

Code	Measure	EnergyPlus objects	Detail
L4-1	LED fixture upgrade — LPD ×0.55 (45 % cut)	Lights.Watts_per_Zone_Floor_Area ×0.55; preserve Return_Air_Fraction , Fraction_Radiant , Fraction_Visible	Exempt: emergency/egress/exit, exterior/facade/parking, theatrical/accent, surgical/cleanroom objects per ASHRAE 90.1-2022 §9.4.1.1
L4-2	Occupancy-sensor schedule trim	Clone Schedule:Compact → _EEM4_OccTrim_ prefix; peak ×0.90, after-hours floor 0.05; skip if Min_Fraction > 0.40 (24/7 loads)	Targets transient zones only; living rooms / bedrooms effectively unaffected at whole-home blend → −5 % Lighting

Equipment-side

Code	Measure	EnergyPlus objects	Detail
E4-1	ENERGY STAR appliance class — EPD ×0.75 (×0.65 BTAP houses)	<code>ElectricEquipment.Watts_per_Zone_Floor_Area</code> (or <code>Design_Level / Watts_per_Person</code>)	Skip process/refrig/kitchen/medical/lab/datacenter objects by Name + EndUse Subcategory regex; BTAP residential override (filename <code>{?i}</code>) (<code>detachedhouse attachedhouse nbc936</code>) suspends commercial skip; Hospital + OutPatient fully skipped
E4-2	Smart power management schedule	Clone Equipment schedule; off-hours floor 0.15 → 0.05; peak ×0.95	Hospital + OutPatient skipped
E4-3	Tier-1 APS phantom cut	Same cloned schedule; sleeping-hours fraction 0.05 → 0.02 (residual standby)	Hospital + OutPatient skipped
E4-4	HEMS behavioural peak trim	Same cloned schedule; daytime/evening peak ×0.96 (4 % cut — Columbia submetering lower bound)	Hospital + OutPatient skipped
E4-5	Heat-Pump Dryer fuel-switch — BTAP residential only	Spawn <code>ElectricEquipment electric_dryer{N}</code> from <code>GasEquipment gas_dryer{N}</code> , <code>Design_Level</code> ×0.50; zero source	Gated: <code>_eem4_is_bt看ap_residential()</code>
E4-6	Induction Range fuel-switch — BTAP residential only	Spawn <code>ElectricEquipment induction_range{N}</code> from <code>GasEquipment gas_range{N}</code> , <code>Design_Level</code> ×0.45; zero source	Gated: <code>_eem4_is_bt看ap_residential()</code>
E4-7	Gas MELs trim — BTAP residential only	<code>GasEquipment gas_mels{N}.Design_Level</code> ×0.85 in-place	Gated: <code>_eem4_is_bt看ap_residential()</code>

Code prefix: **C** = Cooling-side impact · **L** = Lighting-side impact · **E** = Equipment-side impact.

Design principle: Combined moderate-tier measures (not maximum specs) — LED ×0.55 + ESTAR ×0.75 + automated blinds + daylight dimming + occupancy schedules yield cross-load savings (equipment heat-gain → cooling reduction) that exceed any single deep-tier measure alone, at a defensible moderate-investment scenario.

Key guardrails:

- Process load skip (E4-1/2/3/4): regex `{?i}(process|fume.?hood|refrig|walk.?in|kitchen.*equip|cook|med(ical)?.*equip|surg|it.?load|server|datacenter|...)` on Name + EndUse Subcategory.
- Hospital + OutPatientHealthCare: E4-1/2/3/4 fully disabled; C4-1 / C4-3 / L4-1 / L4-2 remain active.
- BTAP residential override (bullet 13): commercial skip suspended so E4-1 fires on `refrigerator1`, `dishwasher1`, etc.; ×0.65 ESTAR floor applied (bullet 15).
- Lab / TallBuilding / SuperTallBuilding: C4-1 (blind injection) skipped due to blank-fenestration-name E+ v22.1 fatal; all other measures active.
- Schedule cloning: every mutated `Schedule:Compact` is cloned with `_EEM4_OccTrim_` / `_EEM4_EquipTrim_` prefix — never mutated in place.
- DataCenter passthrough: no `Lights` or general `ElectricEquipment` → nothing applied.

References:

- **C4-1 automated blinds:** Karlsen, Heiselberg & Bryn (2015) — *Solar shading control strategies for residential buildings in cold climates*; D'Oca et al. (2018) — IEA SHC Task 56 synthesis.
- **C4-3 daylighting:** PNNL-23800 (2014) — *Cost-Effectiveness of 2015 IECC*; Williams et al. LBNL (2012) — *Lighting Controls in Commercial Buildings*.
- **L4-1 LED:** DOE/EE-2136 (2020) — *SSL Adoption Report*; PNNL-23800 40–50 % LPD drop vs pre-LED; Elliott et al. PNNL-32815 (2021).
- **L4-2 occupancy sensor:** Mass EEAC RLPNC 19-2 (2019); CPUC DEER Lighting Controls (2021); NEEP Residential Lighting Strategy (2018).

- **E4-1 ENERGY STAR:** Lobato et al. NREL/TP-5500-49975 (2011) — 25–35 % gain band; EPA ENERGY STAR Unit Energy Consumption Tables (2022).
- **E4-2/3 smart power + APS:** Mass EEAC RLPNC 17-3 (2018); NREL/SR-5500-60000 (2014); PG&E APS Impact Study (2019).
- **E4-4 HEMS:** Allcott (2011) — *Social norms and energy conservation*; Columbia University submetering (2020); EPRI (2018).
- **E4-5/6 fuel-switch:** CEC-500-2021-025; ANSI/RESNET/ICC 301-2022; Energy350 BC Study (2018).

v6 X-cuts and deferrals

Eight measures that were active in v3/v5 were intentionally commented out in v6 to ensure HVAC-tier = ASHP and DHW-tier = HPWH are literally true. All are preserved in the codebase under `# [v6 X-cut]` markers for recovery in a future passive/envelope tier:

Code	Measure	Function(s) commented out
X4	Ventilation HR: HRV/ERV sensible/latent upgrade + missing-HR injection on bare OA paths + zone-ERV fan tune	<code>_enhance_heat_recovery</code> , <code>_add_missing_heat_recovery</code> , <code>_tune_zone_erv</code>
X5	Supply-fan SFP tune ($\eta \geq 0.70$, pressure rise ≤ 600 Pa)	Applied at both EEM2 and EEM3
X6	DHW demand $\times 0.70$ (low-flow fixtures)	<code>apply_dhw_demand_reduction</code>
X7	DHW distribution wrapper (pipe insulation + temp lowering + recirc schedule; central-DHW gated, no-op on SF)	<code>apply_dhw_pipe_insulation</code> , <code>apply_dhw_temp_lowering</code> , <code>apply_dhw_recirc_scheduling</code>
X8	Deep DHW volume right-sizing ($\times 0.50$ SF)	<code>apply_deep_dhw_volume</code>
X9	DHW pipe insulation SF	<code>apply_dhw_pipe_insulation</code>
X10	DHW temp lowering SF (60→55 °C + Sun pasteurize 65 °C)	<code>apply_dhw_temp_lowering</code>
X11	DWHR drain-water heat recovery preheat SF (+8 °C cold-water schedule)	<code>apply_dwhr_preheat</code>

Rationale: X4–X11 are non-HP load-side or distribution measures; their presence in the same EEM tier as ASHP/HPWH conflates heat-pump performance attribution with envelope/distribution efficiency, obscuring the HP-focused contribution claim of the journal paper.

Deferrals to v7 commercial round (separate from X-cuts):

- **A3** (SCWH integrated HP): code-only in v6; predicate fires only for Hotel/Hospital/SuperTall; deferred because the commercial cooling-waste-heat topology is absent from all four residential archetypes.
- **A6/I4a** (hydronic combi HP): code-only in v6.y; predicate gates to radiant-loop commercial buildings; deferred due to commercial radiant-loop topology mismatch.

What is deliberately excluded from all scenarios

- **Internal loads:** No LED lighting upgrades, no plug-load reduction.
- **Scheduling:** No occupancy setback, no demand-controlled ventilation (DCV).
- **HRV/ERV upgrades and missing-HR injection:** Cut from EEM2 in v6 (X4) to maintain HP-only attribution; were active in v3/v5.
- **Supply-fan SFP tune:** Cut from EEM2 and EEM3 in v6 (X5) for the same reason; active in v3/v5. As a consequence, v6 EEM2/EEM3 fan energy ("Other" end use) is approximately 2–3 kWh/m² higher than v3/v5.
- **DHW demand-side and distribution measures (X6–X11):** Cut from EEM3 in v6; active in v3. The removal of X6 (DHW demand $\times 0.70$) raises SF EEM3 DHW by ~30% relative to v5, which is compensated by the E1–E4 HPWH efficiency stack (rated COP 4.0 + CCHP + PLF flatten + VariableSpeed coil).

PV pairing

Scenario	PV treatment
EEM_Default	Native PV only (no Tier-3 override; 0.0 kWh/m² across all Canadian baselines)
EEM1 / EEM2 / EEM3 / EEM4	Tier-3 PV applied. Pitched archetypes (Detached/Attached/OfficeSmall/Restaurants) use the south-facing dominant pitched face only (dominant-face fallback if no south slope); flat roofs use a single world-south 45° rack. No suitability cap. (south-only since 2026-05-29; previously all roof planes)

Net EUI = site gross EUI – annual PV generation (kWh/m²). EUI-ladder analysis uses **gross EUI** to isolate measure effect from PV pairing. PV is reported in dedicated columns and is never netted into EUI for ladder comparisons.

Simulation setup — Canadian cohort (v6)

Item	Value
Total buildings per climate	32 (CAN_MTL CZ6 and CAN_CLG CZ7A run separately)
Building cohort	4 residential archetypes (Detached, Attached, MidRise, HighRise) + 2 tall-residential (ST15, ST20) + NECB17 commercial + OpenStudio supplementals + 4 datacenter (passthrough)
Baseline standard	NECB 2017 (commercial) / NBC 9.36 (residential); BTAP internal loads and schedules
Weather — CZ6	CWEC2020v2 Montréal Trudeau Intl AP
Weather — CZ7A	CWEC2020v2 Calgary Olympic Park Upper
EnergyPlus version	22.1.0
Scenarios per building	5 (Default + EEM1 + EEM2 + EEM3 + EEM4)
Total simulations	320 (32 buildings × 5 scenarios × 2 climates)
DC archetypes	4 per climate (passthrough — EEM1/2/3 rows cloned from Default)
Active buildings in ladder	25 buildings × 2 climates; Datacenter, Supermarket, and OfficeLarge excluded from EUI ladder

EUI ladder — results summary

Climate	Archetype	Default	EEM1	EEM2	EEM3	EEM3 Δ%
CAN_MTL (Z6)	DetachedHouse	127.8	105.3	93.3	81.1	–36.5%
CAN_MTL (Z6)	AttachedHouse	131.0	112.4	106.5	91.9	–29.9%
CAN_MTL (Z6)	AptMidRise	109.2	102.6	95.4	69.1	–36.7%
CAN_MTL (Z6)	AptHighRise	119.9	112.3	110.1	73.0	–39.1%
CAN_CLG (Z7A)	DetachedHouse	116.7	101.0	90.3	78.3	–32.9%
CAN_CLG (Z7A)	AttachedHouse	119.4	107.6	103.5	89.1	–25.4%
CAN_CLG (Z7A)	AptMidRise	103.4	101.2	94.3	68.1	–34.1%
CAN_CLG (Z7A)	AptHighRise	113.3	112.1	110.2	73.2	–35.4%

Fleet median EEM3 Δ% = –34.75% (median of 8 residential data points above). MTL residential median = –36.6%; CLG residential median = –33.5%.

Phase K MidRise DHW note: AptMidRise EEM3 DHW = 17.6 kWh/m² (MTL) / 17.7 kWh/m² (CLG). This is the published, physically defensible value for the NECB17 per-unit OutdoorAirOnly Pumped HPWH topology in CZ6/CZ7A. Phase I+J attempts to converge toward the NEEA/WSU CZ4C reference of ~12 kWh/m² were abandoned in Phase K: (1) that reference describes a centralized CO₂ swing-tank in CZ4C (Pacific Northwest), not the per-unit topology here; (2) the only model lever capable of reaching [12, 16] kWh/m² (forcing :VariableSpeed coils on 23 per-unit MidRise tanks) produced an E+ v22.1 artifact — autosized evaporator fan coupled to coil capacity, with no clean decoupling available.

Phase P note: HighRise EEM3 numbers above are pre-Phase-P. C7 Stratified backup-resistance energy (~7 kWh/m²) is currently mis-bucketed from Water Heating to Space Heating in E+ v22.1. Phase P will add EndUse_Subcategory = 'Water Heating' to both heater objects (label-only fix: combined Htg+WS conserved). See Phase P section below.

Full Option 7i fleet ladder (25 active buildings × 2 climates). Commercial range: –6.6% (Hospital CLG) to –67.7% (Laboratory CLG). Datacenter, Supermarket, and OfficeLarge excluded from ladder analysis.

EEM4 results — residential fleet (v7, 2026-05-13/14)

Disaggregated end-use EUI — residential archetypes (EEM3 → EEM4 delta)

Building	Scenario	Htg	Clg	DHW	Ltg	Equip	Other	Total	Δ% vs Default
Detached (MTL)	EEM3	8.3	7.1	6.7	4.7	46.6	7.8	81.1	–36.5%
Detached (MTL)	EEM4	9.8	5.7	6.7	2.6	28.3	7.8	60.9	–52.3%
Detached (CLG)	EEM3	7.7	5.1	6.6	4.7	46.6	7.6	78.3	–32.9%
Detached (CLG)	EEM4	9.2	4.0	6.6	2.6	28.3	7.7	58.5	–49.9%
Attached (MTL)	EEM3	2.6	7.0	11.2	5.8	57.1	8.1	91.7	–30.0%
Attached (MTL)	EEM4	4.1	5.1	11.2	3.2	34.8	8.0	66.3	–49.4%
Attached (CLG)	EEM3	1.8	5.3	11.1	5.8	57.1	7.8	88.9	–25.5%
Attached (CLG)	EEM4	3.2	3.5	11.2	3.2	34.8	7.7	63.5	–46.8%
MidRise (MTL)	EEM3	4.0	12.8	17.6	4.8	19.3	10.6	69.1	–36.7%
MidRise (MTL)	EEM4	4.3	9.7	17.6	1.9	11.2	9.7	54.4	–50.2%
MidRise (CLG)	EEM3	4.1	11.5	17.7	4.8	19.3	10.6	68.1	–34.1%
MidRise (CLG)	EEM4	4.5	8.6	17.7	1.9	11.2	9.6	53.5	–48.3%
HighRise (MTL)	EEM3	9.7	16.3	8.3	3.9	19.0	15.8	73.0	–39.1%
HighRise (MTL)	EEM4	10.4	11.9	8.3	1.4	10.9	13.9	56.8	–52.6%
HighRise (CLG)	EEM3	9.2	16.1	8.3	3.9	19.0	16.7	73.2	–35.4%
HighRise (CLG)	EEM4	9.9	11.0	8.4	1.4	10.9	14.2	55.8	–50.8%

Fleet median EEM4 Δ% = –50.3% (residential, vs Default). MTL median = –51.3%; CLG median = –48.5%.

EEM3 → EEM4 incremental Δ: Cooling –22 to –33%; Lighting –55 to –64%; Equipment –41 to –43%; Heating +1.5–2.5 kWh/m² (lighting/equipment heat-gain trade-off, fully expected; dominated by Equip + Ltg savings).

Full Option 7i fleet EUI ladder — EEM4 (25 active buildings × 2 climates)

Building	Baseline (MTL)	EEM1 (MTL)	EEM2 (MTL)	EEM3 (MTL)	EEM4 (MTL)	Δ% (MTL)	Baseline (CLG)	EEM1 (CLG)	EEM2 (CLG)	EEM3 (CLG)	EEM4 (CLG)	Δ% (CLG)
Attached Houses	131.0	112.4	106.5	91.7	66.3	-49.4%	119.4	107.6	103.5	88.9	63.5	-46.8%
Detached Houses	127.8	105.3	93.3	81.1	60.9	-52.3%	116.7	101.0	90.3	78.3	58.5	-49.9%
ApartmentHighRise	119.9	112.3	110.0	72.8	56.8	-52.6%	113.3	112.1	110.1	73.1	55.8	-50.8%
HighRise_ST15	113.9	107.2	104.9	70.8	54.6	-52.1%	107.5	107.2	105.4	71.4	53.8	-50.0%
HighRise_ST20	111.0	104.5	102.3	69.0	52.7	-52.5%	104.6	104.7	102.9	69.7	52.0	-50.3%
ApartmentMidRise	109.2	102.6	95.4	69.1	54.4	-50.2%	103.4	101.2	94.3	68.1	53.5	-48.3%
College	81.1	78.3	76.7	63.5	38.0	-53.1%	74.5	73.9	72.6	59.4	34.3	-54.0%
Hospital	265.0	261.8	244.0	229.6	212.0	-20.0%	253.4	249.4	240.5	226.2	207.7	-18.0%
HotelLarge	206.1	199.7	173.9	124.3	114.5	-44.4%	199.3	195.0	171.3	121.9	111.5	-44.1%
HotelSmall	161.6	152.2	142.6	92.4	81.8	-49.4%	155.9	148.3	140.3	90.3	79.8	-48.8%
Laboratory	828.9	787.9	259.7	254.7	229.6	-72.3%	739.1	702.6	232.1	227.1	200.6	-72.9%
MT5 SmallRetail	274.2	173.2	172.1	163.9	157.9	-42.4%	259.5	161.9	161.4	153.2	147.9	-43.0%
OfficeMedium	92.5	82.6	68.5	62.2	40.8	-55.9%	84.8	77.4	64.3	58.0	38.2	-55.0%
OfficeSmall	89.6	81.4	78.5	65.9	50.4	-43.8%	85.4	79.4	76.8	64.2	49.0	-42.6%
OutPatientHealthCare	232.8	216.3	135.6	118.3	110.5	-52.5%	213.3	195.4	126.7	109.3	101.1	-52.6%
RestaurantFastFood	1648.8	1459.0	1064.2	907.4	895.9	-45.7%	1614.1	1422.4	1065.3	908.9	899.6	-44.3%
RestaurantSitDown	1031.0	980.5	670.8	549.4	539.7	-47.7%	1009.4	961.2	677.1	556.5	548.2	-45.7%
RetailStandalone	145.8	130.0	76.2	66.3	53.8	-63.1%	139.1	131.1	79.5	69.6	58.2	-58.2%
RetailStripmall	170.4	135.5	85.5	78.0	57.0	-66.5%	161.0	130.4	85.6	78.2	58.9	-63.4%
SchoolPrimary	143.3	115.4	78.0	71.5	58.3	-59.3%	133.5	114.2	75.4	69.0	55.8	-58.2%
SchoolSecondary	133.8	113.1	72.7	66.6	51.9	-61.2%	120.6	102.9	68.4	62.2	48.9	-59.5%
SuperTallBuilding	109.2	94.3	89.0	70.7	50.6	-53.6%	100.2	91.2	86.0	67.8	47.7	-52.4%
TallBuilding	116.7	105.7	100.5	80.9	59.5	-49.0%	108.8	101.8	96.7	77.1	55.8	-48.7%
Warehouse (50pct)	64.8	59.0	41.4	39.3	37.6	-42.0%	57.5	53.5	34.8	32.7	31.4	-45.4%
Warehouse (full)	47.8	46.1	30.0	28.8	27.2	-43.1%	43.9	44.6	27.6	26.4	24.9	-43.3%

Commercial EEM3 → EEM4 incremental impact by tier:

- **HUGE (-13 to -19 %):** OfficeSmall/Medium, Schools, RetailStandalone/Stripmall — high LPD/EPD baselines, shallow-plan perimeter zones with WWR ≥ 0.20; LED + ESTAR + daylighting all fire together.
- **GOOD (-8 to -12 %):** College, Hotels, Tall/SuperTall, MT5 SmallRetail — mid-tier LPD/EPD or partially deep-plan; LED + ESTAR dominate.
- **MODEST (-4 to -8 %):** Hospital, OutPatient, Warehouses — healthcare equipment skip; low LPD/EPD baselines in Warehouses.
- **NOTHING (< -3 %):** Restaurants, Laboratory — process loads (kitchen/lab/fume hoods) carry > 70 % of EUI; blind/daylighting area is minimal.

Note on Option 9i neighbourhoods: EEM4 not yet simulated at the neighbourhood level — Option 9i EEM4 rerun pending. The neighbourhood ladder below covers EEM1–EEM3 only.

Neighbourhood (Option 9i) results — cumulative ladder and isolated scenarios

Two complementary neighbourhood-scale (Option 9i) EUI views. The **cumulative** ladder applies the EEM tiers as strict supersets ($EEM1 \subset EEM2 \subset EEM3 \subset EEM4$); the **isolated** scenarios apply each domain alone against the Default baseline with HVAC kept on (Envelope-only / HVAC-only / DHW-only / EEM4-only), so each domain's standalone value is separated from sequencing. All values gross site EUI (kWh/m²); PV not netted. (The Option 9i EEM4 rerun noted as 'pending' earlier in this doc completed 2026-05-15 and is included in the cumulative ladder below.)

Option 9i neighbourhood EUI ladder — MTL vs CLG (kWh/m²) — 36 NUs (2026-05-12 / EEM4: 2026-05-15)

$EEM \Delta\% = (EEM4 - Baseline) / Baseline \times 100$. PV not netted into EUI.

Neighbourhood	Baseline (Montreal)	EEM1 (MTL)	EEM2 (MTL)	EEM3 (MTL)	EEM4 (MTL)	EEM Δ% (MTL)	Baseline (Calgary)	EEM1 (CLG)	EEM2 (CLG)	EEM3 (CLG)	EEM4 (CLG)	EEM Δ% (CLG)
RS-I1	169.2	148.6	124.8	100.5	83.4	-50.7%	163.9	147.0	123.6	99.3	82.4	-49.7%
RS-S	202.1	182.1	152.7	130.6	108.0	-46.6%	199.7	179.4	151.1	129.1	106.7	-46.6%
RS-I2	145.4	129.0	101.0	82.3	64.9	-55.4%	136.2	125.9	98.5	79.8	63.5	-53.4%
RS-I3	179.2	144.7	121.3	106.4	85.2	-52.5%	177.0	144.1	119.4	104.6	83.4	-52.9%
RS-I4	137.2	120.3	95.8	76.4	60.6	-55.8%	127.6	115.8	93.2	74.0	59.2	-53.6%
RC-HR1	111.2	103.4	99.8	67.3	52.3	-53.0%	104.4	102.0	99.2	66.7	51.4	-50.8%
RC-HR2	116.2	108.0	106.5	69.0	53.7	-53.8%	109.1	107.1	106.2	68.9	52.8	-51.6%
RC-MR1	113.6	100.5	96.7	74.3	56.4	-50.4%	108.2	98.2	95.0	72.7	54.7	-49.4%
RC-MR2	107.0	99.1	92.4	65.7	51.5	-51.9%	100.2	97.3	91.3	64.7	50.6	-49.5%
RC-MR3	107.0	99.0	92.0	65.3	51.1	-52.2%	100.2	97.2	90.8	64.3	50.2	-49.9%
RC-D	138.6	108.2	93.1	81.0	61.1	-55.9%	135.1	104.3	90.3	78.3	58.9	-56.4%
RC-ML	139.2	111.0	99.7	86.2	63.8	-54.2%	134.3	106.7	96.7	83.4	61.4	-54.3%
RC-T	140.0	114.0	106.5	91.8	66.7	-52.4%	133.4	109.2	103.4	88.8	63.9	-52.1%
RC-R	137.8	107.6	93.2	81.0	61.0	-55.7%	133.5	103.1	90.2	78.2	58.6	-56.1%
RC-R_Garage	137.5	107.8	94.3	90.1	80.9	-41.2%	133.2	103.2	90.9	86.7	77.8	-41.6%
IC-DC	3660.5	3663.7	3645.7	3637.5	3632.7	-0.8%	3654.5	3655.5	3638.2	3630.0	3625.4	-0.8%
IC-DE	11920.8	11914.9	11887.0	11877.1	11865.4	-0.5%	11905.2	11897.1	11868.6	11858.8	11848.4	-0.5%
MU-C1	205.6	182.2	169.2	139.9	119.1	-42.1%	195.3	177.3	167.1	137.9	113.2	-42.0%
MU-C2	136.5	118.0	109.3	85.2	60.2	-55.9%	128.8	116.4	108.5	84.5	58.7	-54.4%
MU-HC	162.6	152.7	124.4	92.6	77.5	-52.3%	154.9	149.1	121.6	89.8	75.0	-51.6%
MU-HS	198.0	176.2	135.3	109.3	94.2	-52.4%	190.6	169.9	132.4	106.4	91.2	-52.2%
MU-L	158.7	141.0	122.8	94.9	80.1	-49.5%	152.3	138.0	121.5	93.7	79.4	-47.9%
MU-S1	193.3	182.2	151.0	122.9	100.5	-48.0%	187.3	182.4	150.0	121.8	100.0	-46.6%
MU-S2	186.8	177.3	148.3	121.7	98.1	-47.5%	180.0	176.2	147.7	121.1	98.0	-45.6%
MU-U1	119.9	100.7	97.3	77.0	50.0	-58.3%	112.0	99.9	97.0	76.7	48.6	-56.6%
MU-W	144.9	129.9	103.9	83.7	74.0	-48.9%	139.2	127.3	102.8	82.6	73.5	-47.2%
MU-W2	128.0	110.7	83.6	73.6	67.4	-47.3%	123.4	107.1	80.8	70.8	65.4	-47.0%
CC-B	119.5	113.6	100.4	87.2	64.5	-46.0%	114.4	109.7	97.5	84.4	62.1	-45.7%
CC-S1	255.4	233.5	190.8	173.2	139.3	-45.5%	245.5	226.1	186.6	168.9	137.2	-44.1%
CC-S2	343.5	316.4	265.3	246.4	196.7	-42.7%	334.0	309.9	261.3	242.2	193.6	-42.0%
CC-E1	95.2	89.3	81.2	71.6	48.8	-48.7%	89.9	86.3	78.7	69.1	46.6	-48.2%
CC-E2	299.6	280.6	145.0	129.7	106.7	-64.4%	273.7	255.7	136.4	121.1	98.9	-63.9%
CC-E3	285.0	269.3	143.8	127.0	106.4	-62.7%	262.4	248.2	135.7	118.9	98.5	-62.5%
CC-FD1	125.0	112.7	101.5	81.8	61.4	-50.9%	116.7	109.6	98.2	78.5	58.5	-49.9%
CC-FD2	117.5	109.6	100.3	84.8	62.5	-46.8%	110.4	106.2	97.4	81.9	59.7	-45.9%
CC-FD3	133.0	119.7	108.1	88.5	66.5	-50.0%	124.6	115.8	104.6	85.2	63.1	-49.4%

Fleet median EEM4 Δ% (36 NUs): MTL = -50.8% · CLG = -49.6%. Fleet median EEM4 EUI: MTL = 67.1 kWh/m² · CLG = 64.6 kWh/m² (36 NUs).
EEM3 reference: MTL = -37.9% / 87.9 kWh/m² · CLG = -35.4% / 84.9 kWh/m².

Isolated EEM scenarios — neighbourhood EUI (CAN_MTL vs US_ASHRAE, kWh/m²) — 35 NUs

Scenario map - SEPARATE (isolated) scenarios, NOT a cumulative ladder

Each EEM domain is applied **on its own** against the DEFAULT baseline, with HVAC kept ON. These are five independent simulations per (NU, standard); the measures are **not** stacked on top of one another. (This differs from the Paper 2 / EEM_Journal cumulative ladder, where EEM2 = EEM1 + HVAC, EEM3 = EEM2 + DHW, etc.) The isolated framing is what lets Block A/B attribute a standalone saving to each domain (and feeds the Shapley decomposition).

Scenario	Flags (env, hvac, dhw, eem4)	scenario_tag	Meaning
Baseline	0, 0, 0, 0	EEM_J_DEFAULT	NECB 2017 / NBC 9.36 baseline, no measures
Envelope	1, 0, 0, 0	EEM_J_ENVELOPE	Envelope retrofit only
HVAC	0, 1, 0, 0	EEM_J_HVAC_ONLY	ASHP / HVAC retrofit only
DHW	0, 0, 1, 0	EEM_J_DHW_ONLY	HPWH / DHW retrofit only
EEM4	0, 0, 0, 1	EEM_J_EEM4_ONLY	Cooling/Lighting/Equipment retrofit only

Each measure cell below shows the isolated EUI and, in parentheses, the standalone change vs that NU's Baseline: Δ% = (measure – Baseline) / Baseline x 100. A negative value is a saving.

Neighbourhood	Baseline (MTL)	Envelope (MTL)	HVAC (MTL)	DHW (MTL)	EEM4 (MTL)	Baseline (US)	Envelope (US)	HVAC (US)	DHW (US)	EEM4 (US)
RS-I1	169.2	148.5 (−12.2%)	128.9 (−23.8%)	145.5 (−14.0%)	158.2 (−6.5%)	190.6	163.5 (−14.3%)	159.2 (−16.5%)	167.7 (−12.0%)	173.9 (−8.8%)
RS-I2	145.3	128.9 (−11.3%)	107.2 (−26.2%)	127.9 (−12.0%)	131.0 (−9.9%)	158.2	132.4 (−16.3%)	131.2 (−17.0%)	140.9 (−10.9%)	142.4 (−10.0%)
RS-I3	179.2	144.7 (−19.2%)	133.4 (−25.6%)	165.1 (−7.9%)	166.2 (−7.3%)	179.1	149.3 (−16.6%)	147.1 (−17.8%)	165.1 (−7.8%)	161.3 (−9.9%)
RS-I4	137.2	120.3 (−12.3%)	101.9 (−25.7%)	119.3 (−13.0%)	130.9 (−4.5%)	153.0	124.3 (−18.8%)	126.2 (−17.5%)	135.3 (−11.6%)	137.6 (−10.1%)
RS-S	202.0	182.0 (−9.9%)	159.8 (−20.9%)	179.4 (−11.2%)	184.4 (−8.7%)	200.0	183.5 (−8.3%)	175.3 (−12.3%)	177.7 (−11.1%)	180.3 (−9.8%)
RC-D	138.6	108.2 (−21.9%)	103.5 (−25.3%)	126.0 (−9.0%)	123.7 (−10.7%)	131.7	103.8 (−21.2%)	99.2 (−24.7%)	119.2 (−9.5%)	116.7 (−11.4%)
RC-HR1	111.2	103.4 (−7.1%)	100.5 (−9.6%)	79.0 (−29.0%)	101.7 (−8.5%)	146.6	120.1 (−18.1%)	125.0 (−14.7%)	114.9 (−21.6%)	132.7 (−9.4%)
RC-HR2	116.2	107.9 (−7.1%)	106.2 (−8.6%)	78.7 (−32.2%)	107.1 (−7.8%)	155.0	125.0 (−19.3%)	131.5 (−15.2%)	117.7 (−24.1%)	142.2 (−8.2%)
RC-ML	139.2	111.0 (−20.3%)	109.3 (−21.5%)	125.3 (−10.0%)	122.4 (−12.1%)	133.5	107.6 (−19.4%)	105.6 (−20.9%)	119.5 (−10.4%)	116.5 (−12.7%)
RC-MR1	113.6	100.5 (−11.5%)	100.0 (−12.0%)	91.4 (−19.6%)	100.6 (−11.5%)	123.3	112.4 (−8.8%)	112.3 (−8.9%)	101.6 (−17.6%)	104.3 (−15.4%)
RC-MR2	107.0	99.1 (−7.4%)	94.3 (−11.8%)	81.0 (−24.3%)	97.2 (−9.1%)	137.7	113.8 (−17.4%)	117.2 (−14.9%)	112.8 (−18.1%)	122.9 (−10.8%)
RC-MR3	107.0	99.0 (−7.5%)	94.0 (−12.2%)	81.0 (−24.3%)	97.4 (−9.0%)	137.9	113.4 (−17.7%)	117.0 (−15.1%)	113.0 (−18.1%)	123.3 (−10.6%)
RC-R	137.8	107.6 (−21.9%)	103.4 (−25.0%)	125.3 (−9.1%)	122.9 (−10.8%)	131.4	103.6 (−21.2%)	99.3 (−24.4%)	118.8 (−9.6%)	116.3 (−11.5%)
RC-T	140.0	114.0 (−18.6%)	115.5 (−17.5%)	124.6 (−11.0%)	121.1 (−13.4%)	135.3	111.5 (−17.6%)	112.3 (−17.0%)	119.9 (−11.4%)	116.2 (−14.1%)
MU-C1	205.5	180.8 (−12.0%)	174.2 (−15.2%)	177.5 (−13.6%)	193.8 (−5.7%)	212.4	188.7 (−11.2%)	188.7 (−11.2%)	184.1 (−13.3%)	199.5 (−6.1%)
MU-C2	136.4	117.0 (−14.2%)	113.0 (−17.1%)	114.8 (−15.8%)	125.2 (−8.2%)	170.0	151.7 (−10.8%)	151.9 (−10.7%)	148.2 (−12.8%)	157.1 (−7.6%)
MU-HC	162.5	152.5 (−6.1%)	125.9 (−22.5%)	131.1 (−19.3%)	152.1 (−6.4%)	194.1	169.1 (−12.9%)	158.0 (−18.6%)	163.1 (−16.0%)	179.9 (−7.3%)
MU-HS	198.0	176.1 (−11.0%)	140.9 (−28.8%)	172.5 (−12.9%)	187.1 (−5.5%)	222.9	196.4 (−11.9%)	179.0 (−19.7%)	197.8 (−11.3%)	211.8 (−5.0%)
MU-L	158.6	140.9 (−11.2%)	131.2 (−17.3%)	131.2 (−17.3%)	148.1 (−6.6%)	178.9	151.2 (−15.5%)	154.1 (−13.8%)	152.5 (−14.7%)	163.9 (−8.4%)
MU-S1	193.0	181.9 (−5.7%)	153.4 (−20.5%)	164.6 (−14.7%)	176.2 (−8.7%)	219.9	191.6 (−12.9%)	184.8 (−16.0%)	192.7 (−12.3%)	200.3 (−8.9%)
MU-S2	186.6	177.2 (−5.0%)	151.1 (−19.0%)	161.0 (−13.7%)	170.4 (−8.7%)	210.8	187.6 (−11.0%)	180.6 (−14.3%)	186.2 (−11.6%)	188.7 (−10.5%)
MU-U1	119.8	99.4 (−17.0%)	101.1 (−15.6%)	101.9 (−14.9%)	107.8 (−10.1%)	154.2	134.7 (−12.7%)	137.9 (−10.6%)	135.7 (−12.0%)	141.2 (−8.4%)

Neighbourhood	Baseline (MTL)	Envelope (MTL)	HVAC (MTL)	DHW (MTL)	EEM4 (MTL)	Baseline (US)	Envelope (US)	HVAC (US)	DHW (US)	EEM4 (US)
MU-W	144.6	129.7 (−10.3%)	111.1 (−23.2%)	125.1 (−13.5%)	138.5 (−4.2%)	161.2	130.6 (−19.0%)	136.9 (−15.1%)	142.1 (−11.8%)	151.3 (−6.1%)
MU-W2	127.5	110.5 (−13.3%)	95.5 (−25.1%)	118.5 (−7.0%)	123.4 (−3.1%)	135.0	98.0 (−27.4%)	117.0 (−13.4%)	126.4 (−6.3%)	130.4 (−3.4%)
CC-B	119.3	113.2 (−5.2%)	100.3 (−15.9%)	106.2 (−11.0%)	101.6 (−14.9%)	181.6	174.2 (−4.1%)	168.6 (−7.2%)	169.0 (−6.9%)	162.6 (−10.5%)
CC-E1	95.1	89.0 (−6.3%)	81.3 (−14.5%)	85.7 (−9.8%)	78.9 (−17.0%)	182.9	160.1 (−12.5%)	158.6 (−13.3%)	173.7 (−5.0%)	164.6 (−10.0%)
CC-E2	299.2	279.2 (−6.7%)	149.4 (−50.1%)	284.0 (−5.1%)	270.1 (−9.7%)	389.5	359.0 (−7.8%)	211.0 (−45.8%)	374.6 (−3.8%)	357.2 (−8.3%)
CC-E3	284.7	269.0 (−5.5%)	145.5 (−48.9%)	268.0 (−5.9%)	262.3 (−7.9%)	376.3	342.8 (−8.9%)	207.8 (−44.8%)	360.2 (−4.3%)	347.8 (−7.6%)
CC-FD1	125.0	112.7 (−9.8%)	101.0 (−19.2%)	105.3 (−15.8%)	110.1 (−11.9%)	188.5	140.4 (−25.5%)	132.4 (−29.8%)	169.3 (−10.2%)	180.6 (−4.2%)
CC-FD2	117.4	109.4 (−6.8%)	99.2 (−15.5%)	101.8 (−13.2%)	100.1 (−14.7%)	170.2	152.3 (−10.5%)	146.0 (−14.2%)	155.1 (−8.9%)	156.5 (−8.1%)
CC-FD3	132.9	119.6 (−10.0%)	108.8 (−18.1%)	113.4 (−14.7%)	116.5 (−12.4%)	167.1	138.4 (−17.1%)	132.1 (−20.9%)	148.0 (−11.4%)	157.4 (−5.8%)
CC-S1	255.0	232.8 (−8.7%)	200.0 (−21.6%)	238.0 (−6.7%)	231.2 (−9.3%)	259.3	233.2 (−10.1%)	225.8 (−12.9%)	242.6 (−6.4%)	235.3 (−9.2%)
CC-S2	342.9	315.6 (−7.9%)	275.3 (−19.7%)	325.2 (−5.2%)	306.7 (−10.5%)	340.6	311.4 (−8.6%)	298.7 (−12.3%)	323.2 (−5.1%)	299.5 (−12.1%)
IC-DC	3,660.2	3,663.5 (+0.1%)	3,636.7 (−0.6%)	3,652.1 (−0.2%)	3,656.6 (−0.1%)	3,679.2	3,670.9 (−0.2%)	3,665.4 (−0.4%)	3,671.5 (−0.2%)	3,674.8 (−0.1%)
IC-DE	11,920.6	11,914.6 (−0.0%)	11,884.0 (−0.3%)	11,910.8 (−0.1%)	11,911.0 (−0.1%)	11,965.4	11,953.3 (−0.1%)	11,943.4 (−0.2%)	11,956.1 (−0.1%)	11,958.0 (−0.1%)

Fleet median isolated Δ% (35 NUs):

Standard	Envelope	HVAC	DHW	EEM4
CAN_MTL	−9.9%	−19.2%	−13.0%	−8.7%
US_ASHRAE	−12.9%	−15.1%	−11.3%	−8.9%

Relationship to v1/v2/v3

v1 (EEM_JOURNAL) applied the same envelope + ASHP bundle as a single tag with no per-tier breakdown; v1 outputs and manuscript-of-record are byte-frozen.

v2 split the bundle into three cumulative tiers with fleet median −30.7% (MTL) / −28.1% (CLG). **v3** deepened to a heat-pump focus, adding HRV/ERV upgrades (X4) and supply-fan SFP (X5) to EEM2, and nine DHW measures (HPWH, DWHR, demand reduction, distribution) to EEM3, reaching −40.6% (MTL) / −40.9% (CLG). **v2 and v3 are superseded by v6.**

Path from v3 to v6:

- **v4 SF-targeted (2026-05-08):** Added E2-J Branch A (per-zone PTHP for Detached/Attached SF), E3-J (HPWH tank jacket), and several SF-focused EEM3 corrections.

- **v5 SF push-forward (2026-05-08/09/10):** Continued E2-J Branch A + HPWH efficiency items (E2-I inverter cooling curves); commercial measures l4a/b/c predicate-gated and inert on residential fleet.
 - **v6.x (2026-05-10):** 13 physics bug fixes (C1–C13, G2–G4); 8 non-HP measures commented out (X4–X11); MURB swing-tank (A1+A2) and SF preheat-tank (A5) topology additions; HPWH rated COP upgraded 2.69 → 4.0 (Phase L — transcritical CO₂, Sanden GAU / Mitsubishi Q-Ton); C7 Mixed→Stratified HPWH tank conversion added.
 - **Phase K (2026-05-10) — MidRise convergence ABANDONED:** Four consecutive Phase I+J pre-flight gate failures drove the decision. **Binding outcome: v6.y MidRise EEM3 DHW = 17.6 kWh/m² is the published, physically defensible value.** v6.y equals v6.x H4 within simulation noise; fleet median unchanged at –34.75%.
 - **Phase P (in flight, 2026-05-11):** Rank 1 labeling-bug fix for C7 Stratified `EndUse_Subcategory` ; will refresh HighRise EEM3 Htg/DHW split in the report once the 224-sim full-fleet re-run completes. Combined Htg+WS conserved; fleet median not expected to change.
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